

Coordinate Geometry

The distance formula and the section formula for internal division, including the midpoint, with worked examples and practice questions.

Introduction

In Class 9 you learned how to plot points on a coordinate plane and how to locate a point given its coordinates. Now we use coordinates to find the distance between two points and the point that divides a line segment internally in a given ratio. These are the two topics listed for CBSE 2026-27.

Distance Formula

Given two points on the plane, we want the straight-line distance between them, the length of the line segment joining them.

Distance Formula

The distance between two points $P(x_1, y_1)$ and $Q(x_2, y_2)$ is

$$PQ = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2}$$

It comes from applying the Pythagoras Theorem to the right triangle formed by the horizontal distance $|x_2 - x_1|$ and the vertical distance $|y_2 - y_1|$ between the points.

A few points to keep in mind:

- The formula works whichever point you call (x_1, y_1) and whichever you call (x_2, y_2) , swapping them gives the same answer because we square the differences.
- The distance is always non-negative. Take the positive square root.
- The distance of a point $P(x, y)$ from the origin $O(0, 0)$ is a special case:

$$OP = \sqrt{x^2 + y^2}$$

Worked Example

Find the distance between A(2, 3) and B(6, 6).

$$AB = \sqrt{(6 - 2)^2 + (6 - 3)^2} = \sqrt{4^2 + 3^2} = \sqrt{16 + 9} = \sqrt{25} = 5$$

This is another appearance of the 3-4-5 right triangle, the differences in x and y (4 and 3) are the legs of the right triangle.

Worked Example

Check whether the points A(1, 2), B(4, 6) and C(7, 2) form an isosceles triangle.

Find all three side lengths.

$$AB = \sqrt{(4-1)^2 + (6-2)^2} = \sqrt{9 + 16} = 5$$

$$BC = \sqrt{(7-4)^2 + (2-6)^2} = \sqrt{9 + 16} = 5$$

$$CA = \sqrt{(7-1)^2 + (2-2)^2} = \sqrt{36 + 0} = 6$$

Since $AB = BC = 5$, the triangle is **isosceles** with $AB = BC$.

Worked Example

Find the point on the x-axis that is equidistant from A(2, -5) and B(-2, 9).

Any point on the x-axis has the form P(x, 0). Equidistant means $PA = PB$, so $PA^2 = PB^2$:

$$PA^2 = (x - 2)^2 + (0 + 5)^2 = (x - 2)^2 + 25$$

$$PB^2 = (x + 2)^2 + (0 - 9)^2 = (x + 2)^2 + 81$$

Setting them equal:

$$(x - 2)^2 + 25 = (x + 2)^2 + 81$$

$$x^2 - 4x + 4 + 25 = x^2 + 4x + 4 + 81$$

$$-4x + 29 = 4x + 85 \implies -8x = 56 \implies x = -7$$

So the required point is **(-7, 0)**.

The distance formula is also the tool for proving that a given set of points forms a particular type of triangle or quadrilateral, just calculate the side lengths (and diagonals for quadrilaterals) and compare them.

Section Formula

Suppose you have two points A and B, and a point P on the line segment AB that divides it in a given ratio. The section formula tells you the coordinates of P without drawing a graph.

Section Formula (Internal Division)

The coordinates of the point $P(x, y)$ which divides the line segment joining $A(x_1, y_1)$ and $B(x_2, y_2)$ internally in the ratio $m_1 : m_2$ are

$$P = \left(\frac{m_1x_2 + m_2x_1}{m_1 + m_2}, \frac{m_1y_2 + m_2y_1}{m_1 + m_2} \right)$$

Here, m_1 is the part next to B and m_2 is the part next to A, think of m_1 as multiplying the coordinates of B and m_2 as multiplying the coordinates of A.

An important special case is the midpoint, where $m_1 = m_2 = 1$:

Midpoint Formula

The midpoint of the segment joining $A(x_1, y_1)$ and $B(x_2, y_2)$ is

$$M = \left(\frac{x_1 + x_2}{2}, \frac{y_1 + y_2}{2} \right)$$

Note that the section formula as given in Class 10 deals with internal division, the point P lies between A and B. The formula works for the ratio counted as $AP : PB = m_1 : m_2$.

Worked Example

Find the coordinates of the point that divides the segment joining $A(2, -3)$ and $B(5, 6)$ internally in the ratio $1 : 2$.

Here $m_1 = 1$, $m_2 = 2$, $A = (2, -3)$, $B = (5, 6)$.

$$x = (1 \cdot 5 + 2 \cdot 2)/(1+2) = (5 + 4)/3 = 9/3 = 3$$

$$y = (1 \cdot 6 + 2 \cdot (-3))/(1+2) = (6 - 6)/3 = 0/3 = 0$$

So the point is **(3, 0)**.

Worked Example

Find the ratio in which the line segment joining $A(1, -5)$ and $B(-4, 5)$ is divided by the x-axis. Also find the coordinates of the point of division.

Any point on the x-axis has y-coordinate 0. Let the point be P and let $AP : PB = k : 1$. Using the section formula for the y-coordinate:

$$y = (k \cdot 5 + 1 \cdot (-5))/(k+1) = (5k - 5)/(k+1)$$

Set $y = 0$: $5k - 5 = 0 \implies k = 1$. So the x-axis divides the segment in the ratio **1 : 1** (it bisects it).

The x-coordinate is then:

$$x = (1 \cdot (-4) + 1 \cdot 1)/(1+1) = (-3)/2 = -1.5$$

So the point of division is **(-3/2, 0)**.

A useful tip: when you need to find the ratio in which a point divides a segment, assume the ratio is $k : 1$, write the section formula, and solve for k using the known coordinate.

Summary

- **Distance formula:** distance between $P(x_1, y_1)$ and $Q(x_2, y_2)$ is $\sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2}$. Distance from the origin: $\sqrt{x^2 + y^2}$.
- **Section formula (internal):** point dividing $A(x_1, y_1)B(x_2, y_2)$ in the ratio $m_1 : m_2$ is $((m_1x_2 + m_2x_1)/(m_1+m_2), (m_1y_2 + m_2y_1)/(m_1+m_2))$. Midpoint = $((x_1+x_2)/2, (y_1+y_2)/2)$.

Practice Questions

1. Find the distance between the points $P(-5, 7)$ and $Q(-1, 3)$. Also find the distance of the point $(4, 3)$ from the origin.

2. Show that the points $(1, 7)$, $(4, 2)$, $(-1, -1)$ and $(-4, 4)$ are the vertices of a square.

3. Find the coordinates of the point which divides the line segment joining $A(-2, 2)$ and $B(2, 8)$ in the ratio $3 : 1$.

4. Find the ratio in which the point $(-3, p)$ divides the line segment joining the points $(-5, -4)$ and $(-2, 3)$. Also find the value of p .